Deposit Beta Data

Goal: Estimate deposit betas. A deposit beta is how the interest rate banks pay on deposits change when other interest rates change. Here we are focusing on the federal funds rate and the interest rate that changes and then looking at how banks change the interest rates they pay on average. A deposit beta is basically an elasticity.

$$\frac{\Delta deposit\ rate}{\Delta fed\ funds\ rate}$$

Deposits are usually "sticky" as people don't change banks very often and so we would expect deposit betas to be relatively low or inelastic. I believe how the variables are defined in the dataset if we get an estimated beta = 0.25 that would mean a 1 percentage point increase in the federal funds rate increases the average deposit rate at a bank by 0.25 percentage points.

Key Definitions

$\Delta deposit\ rate$ this is the dependent variable in our regressions.

$\Delta fed\ funds\ rate$ this is the key independent variable in our regressions

The deposit rate itself is $\frac{interest\ expense}{deposits}$

We then will have an additional set of control variables. These are variables related to the bank that could be important to the relation ship we are interested in. They are usually entered as one quarter lags, which has already been constructed in the database.

Bank Size: the total assets of the bank
NIB Ratio: non-interest bearing deposits divided by deposits
ROA: net income divided by assets
Leverage Ratio: bank capital divided by assets
Loan Ratio: loans divided by assets
Liquidity Ratio: (cash + funds due from banks + securities) divided by assets
[Note: smaller banks in particular often have funds deposited in bigger banks which is the "funds due from banks" part].

Dataset Information

There are 10 years of data on a quarterly basis for thousands of banks. This is a panel dataset of about 167,560 obersvations. Of course for the change in deposit rate and change in federal funds rate, we will be missing the first quarter of 2016 since that can't be differenced. The same thing is true for one quarter lag variables. Key variables have been "winsorized". That is due to outliers. In datasets like this it is common that a number gets entered incorrectly in some way (e.g. wrong units or wrong column or just 0 entered incorrectly) by the reporting bank. These can be extreme so winsorizing is a way to correct that. In this case the observations below 1 percentile are replaced by the 1st percentile value and observations above 99 percentile are replaced by the 99th percentile. This is a fairly minimal cutting as sometimes people will do 5 and 95.

Variable Names

IDRSSD: the unique number that identifies each bank

date: end date for the quarter the data was collected. For example 03/31/2016 means the end of the first quarter of 2016

year: The year the data was collected
qtr: the quarter of the year the data was collected (e.g. 1, 2, 3, or 4)

depositrate: This is the annualized deposit rate paid by each bank. Since it is quarter, the annualization is to multiply by 400 to make it an annual percentage.

depositratechange: This is the one period change in *depositrate*. This is the dependent variable.

ffrchange: This is the change in the federal funds rate. This is the main independent variable.

lassets: this is the log of bank total assets. Assets were in $1000s of dollars so if *lassets* = 11, then the bank's assets are $e^{11} = 59{,}874.14$ or about $60 million.

lassetslag: this is the one quarter lag of *lassets*. The lag variables will be the control variables in regressions.

nib: non-interest bearing deposits ratio

niblag: one quarter lag of non-interest bearing deposits ratio

roa: return on assets

roalag: one quarter lag of return on assets

leverage: leverage ratio

leveragelag: one quarter lag of leverage ratio

loan: loan to assets ratio

loanlag: one quarter lag of loan to assets ratio

liquidity: liquidity ratio

liquiditylag: one quarter lag of liquidity ratio